

Retrograde Mars (20 points)

The projector will display Mars moving relative to the background stars over one season of visibility (1.5 years) starting from the heliacal rising, chosen so that Mars will be at maximum ecliptic latitude at opposition.

The ecliptic will also be displayed, marked with the positions of the Sun during the year and the current date. The Sun will always be below the horizon.

Synodic period of Mars = 780 days.

(Projection time 10 minutes)

(a) Record the following quantities:

2.a (8 pt)

i.	the dates of quadrature (when the elongation of Mars is 90°)	
ii.	the date of the beginning of retrograde motion	
	and the date of the end of retrograde motion	
iii.	the date of opposition	
iv.	the ecliptic latitude at opposition	
v.	the width in ecliptic longitude of the loop made by the planet	

Based on your observations and assuming the orbits of Earth and Mars are circular,

(b) On the answer sheet, mark the positions of the Sun, Earth and Mars at the moments of opposition and quadrature in the heliocentric system and determine the radius of the orbit of Mars in a.u. geometrically, without using Kepler's Laws. Show your method in the answer sheet.

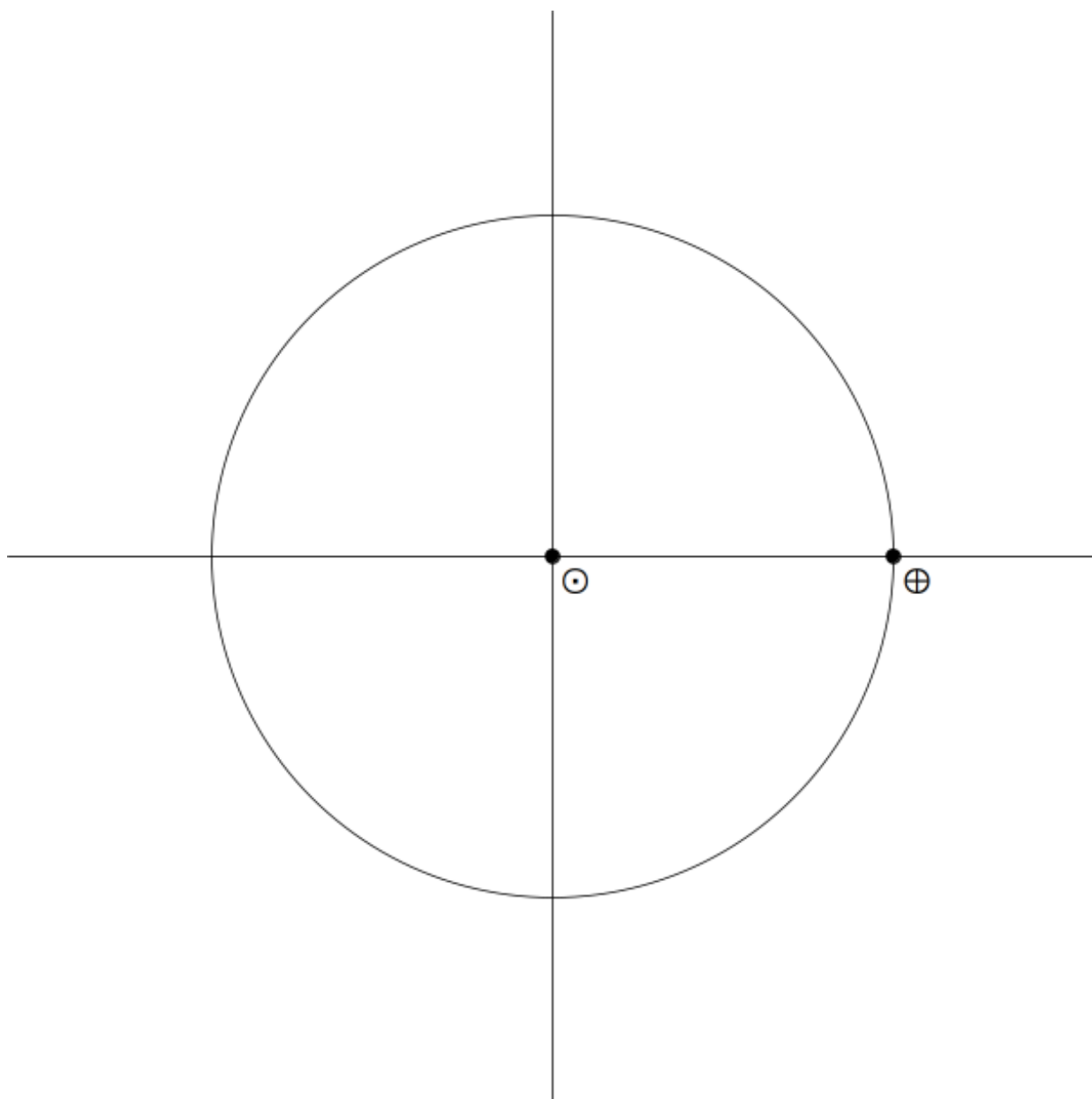
(9 points)

(c) Derive the inclination of the orbit of Mars to the ecliptic.

(3 points)



2.b (9.0 pt)



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Determination of the radius of the orbit of Mars:

2.b (cont.)

Derivation of the inclination of the orbit of Mars to the ecliptic:

2.c (3 pt)